BioX AAL-Band GUI User Guide

Complete guide for sensor data collection and visualisation

BioX ApS

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BioX AAL-Band GUI
User Guide

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Application Overview

The **BioX AAL-Band GUI** is a desktop application for Windows that connects to AAL-Band sensor devices over **Bluetooth**, streams real-time IMU and FSR data, and saves recordings to CSV files for offline analysis.

Interface Structure

The main window contains a single top-level tab—**Bluetooth**—with four sub-tabs:

Sub-Tab	Purpose
Connect	Device discovery, connection, calibration, and status log
Collect	Data recording, CSV file management, and offline plotting
Heatmap	Real-time FSR pressure visualisation
Graphs	Real-time IMU and FSR waveform graphs with battery indicator

Table 1: Sub-tabs available in the Bluetooth top-level tab

Menu Bar

Menu	Item	Action
Flash Firmware	`Flash AALBand Firmware`	Opens the firmware flash tool window
Theme	`Light`	Switch to light colour scheme
Theme	`Dark`	Switch to dark colour scheme
Help	Documentation	Opens BioX documentation website
Help	Updates	Opens BioX downloads page
Help	About Us	Opens BioX company page

Table 2: Menu bar options

Device Firmware

The AAL-Band ships with Bluetooth firmware pre-installed. This user guide covers Bluetooth operation.

Firmware	Connection	GUI Tab
Bluetooth (default)	Bluetooth RFCOMM / Serial Port Profile	Bluetooth

Table 3: AAL-Band firmware variants

Flashing Firmware via the GUI

The GUI includes a built-in firmware flash tool that allows you to update the AAL-Band firmware directly over USB — no external software required. Each release of the GUI ships with the matching firmware version, so the device and software are always kept in sync.

2.1.1 Requirements

- A **USB cable** with a CP210x USB-to-Serial chip (the standard AAL-Band programming cable)

- The **CP210x USB-Serial driver** — installed automatically by the GUI setup installer. If running from source, install it manually from the `drivers/` folder in the project directory

> **Note:** The USB cable used for flashing is **separate** from the Bluetooth connection. Disconnect the device from Bluetooth before flashing.

2.1.2 Flashing Procedure

1. Connect the AAL-Band to your computer using the USB programming cable
2. In the menu bar, click **Flash Firmware** → **Flash AALBand Firmware**
3. The **Firmware Flash Tool** window opens
4. In the **Port** dropdown, select the COM port corresponding to the AAL-Band. CP210x devices are listed first and are identified by their description
   - If no port appears, click **Refresh** and ensure the USB cable is securely connected
5. Click **Flash Firmware**
6. The progress bar animates through the following stages:

Stage	Description
Connecting	Breathing animation while establishing communication with the chip
Chip detected	Device identified; flash memory is being erased
Writing	Four firmware binaries are written sequentially; progress bar fills from left to right
Verifying	Written data is verified against the source files; bar crawls slowly to 99%
Resetting	Device resets and boots the new firmware
Complete	Bar turns solid green; **Done** button appears

Table 4: Firmware flash stages

6. When the status reads **Flash complete!** , click **Done** to close the window
7. Disconnect the USB cable and reconnect the device via Bluetooth as normal

2.1.3 If the Flash Fails

- The progress bar turns red and the status reads **Flash FAILED**
- Check that no other application (e.g. Arduino IDE Serial Monitor) has the COM port open
- Ensure the USB cable is data-capable (not charge-only)
- Click **Flash Firmware** again to retry — the tool resets automatically
- If the problem persists, contact BioX support — see Section 11

Bluetooth Device Connection

Step 1 — Pair the Device with Windows

Before the GUI can connect, the AAL-Band must be paired at the operating-system level:

1. Open **Windows Settings** → **Bluetooth & Devices** → **Add Device**
2. Select your **AAL-Band** from the list
3. Wait until the status shows *Paired / Connected*
4. After the pairing process, the **Reset** button must be pressed at each connection.

Step 2 — Open the Bluetooth Connect Tab

1. Click the **Bluetooth** top-level tab
2. Click the **Connect** sub-tab

The tab contains:

- Device table with columns: **Port**, **Name**, **MAC Address**
- **Search** / **Stop Search** button
- **Calibrate** button
- **Disconnect** button
- System log area (timestamped status messages)

Step 3 — Search for Devices

1. Click **Search**
   - Button text changes to **Stop Search**
   - The GUI scans all available COM ports (typically takes 10–30 seconds)
   - Progress and results are shown in the system log
2. Each successfully connected device appears automatically in the table
3. System log confirms: `Found device on COMX`

Step 4 — Rename Devices (Optional)

1. Double-click the **Name** cell of a device row
2. Type a descriptive name (e.g. *Left Arm*, *Right Arm*)
3. Press **Enter** to save — the name updates across all sub-tabs automatically

Step 5 — Disconnect / Reconnect

- **Disconnect**: Select a row → click **Disconnect**. The device is removed from the table.
- **Reconnect**: Click **Search** again. Paired devices reconnect automatically.

Device Calibration

Calibration zeros the FSR baseline so that readings reflect actual applied pressure rather than sensor offset.

Calibration Procedure

1. Navigate to the **Bluetooth** tab → **Connect** sub-tab
2. Click on the device row to select it
3. Close your fist tightly and hold it
4. Click **Calibrate** — the button changes to **Calibrating...** and becomes disabled
5. Hold your fist closed for **5 seconds**
6. The system log confirms completion; the button returns to **Calibrate**
7. Release your fist — the device is now calibrated.

When to Calibrate

- Upon first connecting a device in a new session
- When FSR readings appear inaccurate or drifted
- After the device has been idle for an extended period

Tip:

- All 8 FSR sensors calibrate simultaneously in a single 5-second procedure.
- The BNO055 IMU has internal calibration algorithms, so there is no need for external calibration methods.

Data Collection and Recording

Starting a Recording Session

1. Click the **Bluetooth** tab → **Collect** sub-tab
2. Select the target device from the **Select device** dropdown
3. (Optional) Enter a custom prefix in the **File name** field. If left blank, a timestamp is used automatically
4. Click **Collect**
   - Button text changes to **Stop**
   - Data streams in real-time to a CSV file
   - Filename format: `Data-<Type>-YYYY.MM.DD-HH.MM.SS.csv`
5. Perform the activity to be recorded
6. Click **Stop** — the file is finalised and appears in the file list below

Saved Data Location

All CSV recordings are saved to:

`Documents\BioX AAL-Band GUI\Saved Data\`

The folder is created automatically on first launch. No administrator rights are required.

CSV File Management

The file list beneath the recording controls shows all saved recordings, sorted by most recent first. Each entry displays the filename, file size, and modification date.

Button	Action
Collect / Stop	Start or stop recording
Plot Selected File	Open the plotting window for the selected file
Delete Selected File	Permanently delete the selected file (confirmation required)
Refresh File List	Re-scan the Saved Data folder and update the list

Table 5: Collect tab controls

Plotting Recorded Data

1. Select a file from the list (click to highlight)
2. Click **Plot Selected File**
3. A separate **Data Plotting** window opens showing all sensor channels
4. Use the navigation buttons to cycle through the 5 graph groups:

Button	Action
Previous Graph	Go to the previous graph group
Next Graph	Go to the next graph group
Graph X/5 indicator	Shows current position within the 5 groups

Table 6: Plotting window navigation

The 5 graph groups are: FSR, Gyroscope, Linear Acceleration, Quaternion, and Gravity.

Real-Time Visualisation

Heatmap Tab

The Heatmap tab provides a live FSR pressure map overlaid on an anatomical diagram of the insole sensor layout.

6.1.1 Accessing the Heatmap

1. Connect a Bluetooth device
2. Click the **Heatmap** sub-tab
3. Select the device from the **Select Device** dropdown at the top
4. The heatmap updates automatically at approximately 10 FPS

6.1.2 Channel Type

Use the **Channel Type** dropdown to switch between two visualisation layouts:

Type	Description
Plain	8 rectangular regions (FSR0–FSR7) arranged vertically along the insole
Round	8 polygon regions (FSR0–FSR7) mapped around a circular insole diagram

Table 7: Heatmap channel types

6.1.3 Colour Scale

Each FSR region is coloured according to its current voltage (0–3.1 V) using the *jet* colourmap:

Colour	Meaning
Blue	No / minimal pressure (0 V)
Cyan	Low pressure
Green	Moderate pressure
Yellow	Medium-high pressure
Red	Maximum pressure (3.1 V)

Table 8: Heatmap colour scale

Each region also displays its FSR index and current voltage value as a text label.

Graphs Tab

The Graphs tab shows five scrolling real-time waveforms, one per sensor group.

6.2.1 Accessing the Graphs

1. Connect a Bluetooth device
2. Click the **Graphs** sub-tab
3. Select the device from the **Select device** dropdown at the top
4. Graphs update automatically (display window: last 3 seconds)

6.2.2 Graph Groups

Graph	Channels	Y-Axis Range	Unit
FSR	FSR0–FSR7 (8 lines)	0 – 5.2	V
Linear Acceleration	X, Y, Z	−50 – +50	m/s^2
Gravity	X, Y, Z	−9.9 – +9.9	m/s^2
Gyroscope	X, Y, Z	−100 – +100	deg/s
Quaternion	W, X, Y, Z	−1 – +1	(normalised)

Table 9: Real-time graph groups and axis ranges

Each graph includes a colour-coded legend in the title row. All graphs are displayed in a vertically scrollable area.

6.2.3 Battery Indicator

A battery progress bar is located in the top-right corner of the Graphs tab. It updates with every incoming data packet.

Level	Colour	Range
Full	Dark green	75–100%
Good	Light green	50–75%
Low	Orange	20–50%
Critical	Red	0–20%

Table 10: Battery indicator colour thresholds

A warning label appears next to the bar when:

- Battery $\leq$ 15%: **Low Battery**
- Battery $\leq$ 5%: **CRITICAL!**

Multi-Device Operation

The GUI supports simultaneous connection to multiple Bluetooth devices.

Setup

- **Bluetooth**: Pair all devices with Windows first, then run Search once — all paired devices are found and connected in one scan.

After connecting, rename each device from the Connect tab for easy identification (e.g. *Left Foot*, *Right Foot*).

Switching Between Devices

All connected devices appear in the dropdown selectors on the Collect, Heatmap, and Graphs sub-tabs. Select the desired device from the dropdown to view or record its data.

Limitations

- Only one device can be *recorded* at a time
- Each additional connected device increases CPU and memory usage
- Real-time graphs display one device at a time; switch using the dropdown

Appearance

Theme

The application supports Light and Dark themes, selectable from the **Theme** menu in the menu bar. The theme affects:

- Application background and text colours

- Graph backgrounds (white in Light, dark grey in Dark)
- Heatmap figure background and label colours
- Windows 10/11 title bar colour (matches the selected theme automatically)

The selected theme is applied immediately to all open windows.

Tips and Best Practices

Before Recording

- Test the device connection and verify that graphs show live data
- Calibrate the device if FSR readings appear inaccurate
- Assign descriptive names to devices before starting
- Check the battery level in the Graphs tab

During Recording

- Keep the GUI window open — closing it stops data collection
- Do not disconnect the device while recording
- Note any anomalies or events of interest during the session

After Recording

- Open the Collect tab and click **Plot Selected File** to verify data quality
- Back up the `Saved Data` folder regularly for important sessions
- Delete test or invalid recordings to keep the file list manageable

Performance

- Limit the number of simultaneously connected devices if the application feels slow
- Heatmap and graph rendering refreshes at 10 FPS to balance responsiveness and CPU load
- Charge devices before long recording sessions

Getting Help

If you encounter issues not covered in this guide:

1. Check the **system log** in the Connect tab for timestamped error messages
2. Visit the **Help** menu → **Documentation** for the latest online resources
3. Contact BioX support — see Section 11 — and provide:
    - The exact error message from the system log
    - Steps to reproduce the issue
    - Screenshots of the GUI state
    - Your Windows version

Contact BioX

For support, firmware requests, bug reports, or general enquiries, please reach out to the BioX team:

Channel	Details
Website	https://www.bioxgroup.dk/contact-biox-group/
Company	BioX ApS

Table 11: BioX contact information

When contacting support regarding a firmware issue, please include:

- Your device serial number or MAC address (visible in the Connect tab device table)
- The firmware currently installed (Bluetooth), if known
- A description of the problem and any error messages from the system log

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